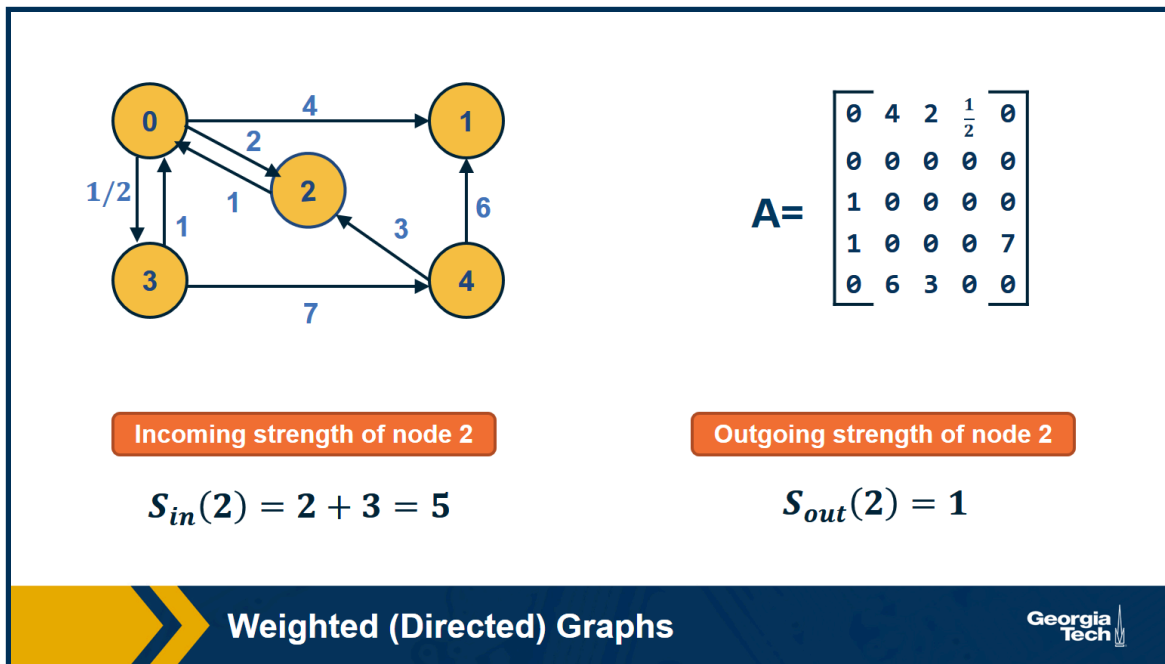


## L2: Weighted Directed Graphs

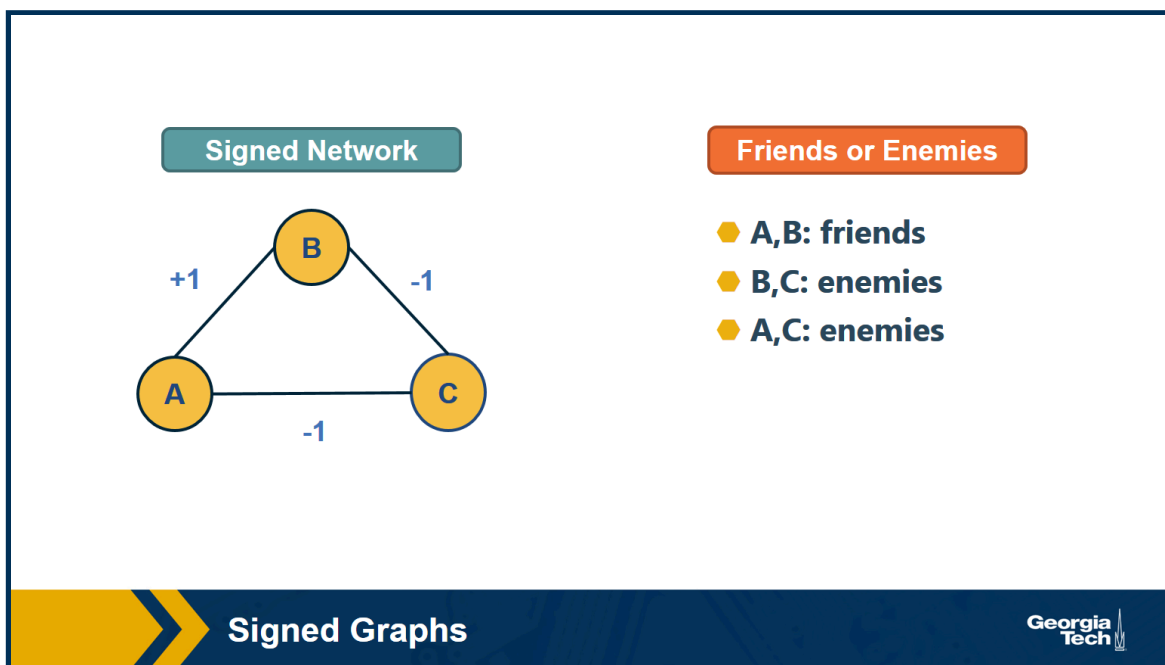
So far we assumed that all edges have the same **strength** – and the elements of the adjacency matrix are either 0s or 1s. In practice, most graphs have edges of different strength – we can represent the strength of an edge with a number. Such graphs are referred to as weighted.

In some cases the edge weights represent **capacity** (especially when there is a flow of some sort through the network). In other cases edge weights represent distance or cost (especially when we are interested in optimizing communication efficiency across the network).

In undirected networks, the “strength” of a node is the sum of weights of all edges that are adjacent to that node.



In directed networks, we define **in-strength** (for incoming edges) and **out-strength** (for outgoing edges).



In ***signed graphs***, the edge weights can be negative, representing competitive interactions. For example, think of a network of people in which there are both friends and enemies (as shown in the visualization above).